

The Navier–Stokes workbench

The exact radius of the Rindler shear series, and curvature rates for the map to Yang–Mills

A paper prepared from an audit of the workbench by Claude (Anthropic),
model `claude-opus-5-5`

Workbench: KokunoYumeto (KokunoYumeto/yang-mills-interacting-workbench, folder
`navier-stokes/`);

imported construction: OpenAI, *Finite Time Blowup for Navier–Stokes*

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condensed record; Appendix B)

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and its later versions

Abstract

OpenAI’s manuscript *Finite Time Blowup for Navier–Stokes* constructs, for every viscosity, a smooth solution of the forced three-dimensional Navier–Stokes equations with bounded energy that blows up in finite time. A workbench built by KokunoYumeto with AI systems reconstructs the construction, continues it into a vacuum-hydrodynamics model, and connects it to Yang–Mills and other programmes. This paper states what the workbench establishes and proves three kinds of results.

First, for the shear sector of the cutoff Rindler fluid of the continuation, the hydrodynamic dispersion series $\omega(k) = -i\nu k^2 - i(3\nu^3/2c^2)k^4 - \dots$ has radius of convergence exactly $k_* = 0.389236400495165 \dots c/\nu$. The radius is set by a certified collision of the diffusive pole with the first non-hydrodynamic pole at purely imaginary frequency; the mode is purely damped up to it, the coefficients have exact square-root asymptotics, and the series summed on its circle of convergence returns the collision frequency. These statements are proved with computer assistance in ball arithmetic and verified independently with a second interval library. Second, conditionally on four named statements of the manuscript, the curvature masses of the workbench’s map from Navier–Stokes to Yang–Mills grow at explicit rates. Third, under the same statements the blowup is of Type II, with $\|u\|_{L^3} \gtrsim \tau^{-4h/3}$, and every rate is compared with the classical necessary conditions for a singularity. The manuscript’s main theorem is imported and is not verified here.

Introduction

The problem and this record

OpenAI’s manuscript [1] is a 166-page construction of finite-time blowup for the forced Navier–Stokes equations in three dimensions. The workbench around it transcribes the manuscript, adds a 208-page reader and programs that replay its identities, and continues it in two directions: into a vacuum-hydrodynamics model built on the cutoff Rindler fluid of fluid/gravity duality, and into typed maps to the author’s other programmes, among them Yang–Mills. Two questions about this material are of independent interest and can be settled by computation: whether the hydrodynamic expansion of the continuation’s Rindler fluid converges, and how the manuscript’s profile estimates translate into rates for the quantities that the map to Yang–Mills uses. This paper settles the first unconditionally and the second conditionally, re-runs every identity of the workbench’s first-pass scripts, and states what each map carries.

The workbench is research carried out with AI systems, which its author used to formulate and test many hypotheses quickly. Some of its directions are finished computations, some are programmes

with proved parts, and some are proposals; not every direction is expected to hold. An unfinished programme is still mathematics, and its proved parts stand on their proofs whatever becomes of the rest. The audit behind this paper had limited compute and did not examine every direction; where the paper gives a view rather than a proof, the view is labelled as mine and its reasons are given.

Main results

The continuation's cutoff Rindler fluid has, for shear perturbations $\propto e^{-i\omega t + ikz}$, the dispersion relation $I'_\alpha(q) = 0$ with $q = k\rho_c$, $\alpha = -i\omega\rho_c/c$ and $\rho_c = 2\nu/c$ (Section 2). The hydrodynamic root α_h is the zero that tends to 0 as $q \rightarrow 0$; in $x = q^2/4$ it has the expansion $\alpha_h(x) = -2x - 3x^2 - \frac{29}{3}x^3 - \dots$.

Theorem I (Theorems 2.1 and 2.2). *The function $(\alpha, q) \mapsto I'_\alpha(q)$ has a nondegenerate double zero in α at (α_*, q_*) with $q_* = 0.778472800990330076180356446189 \pm 10^{-20}$ and $\alpha_* = -0.5697140809723617 \dots$. The hydrodynamic root α_h is analytic on the disk $|x| < x_* = q_*^2/4$, real and negative on $0 < x < x_*$, and has a square-root branch point at x_* . So its Taylor series has radius of convergence exactly x_* ; in physical units, $\omega(k)$ has radius exactly $k_* = q_*c/(2\nu) = 0.389236400495165038 \dots c/\nu$, and its first singularity is $\omega_* = -0.2848570405 \dots i c^2/\nu$.*

Theorem II (Corollary 2.3). *Write $\alpha_h(x) = \sum_{k \geq 1} a_k x^k$. Then $a_k = -C x_*^{-k} k^{-3/2} (1 + O(1/k))$ with a certified constant $C \in [0.14696, 0.14757]$. The series converges absolutely on $|x| = x_*$, and $\sum_k a_k x_*^k = \alpha_*$: the dispersion series summed at $k = k_*$ returns the collision frequency.*

The Yang–Mills repository maps a velocity field u to the abelian connection $A_i = \lambda u_i T$ and uses the curvature masses $\mathcal{K}_B = \lambda^2 \|\text{curl } u\|^2$ and $\mathcal{K}_E = \lambda^2 c^{-2} \|\partial_t u\|^2$. With $\tau = 1 - t$ and the manuscript's small parameter $0 < h < 1/100$:

Theorem III (Theorem 3.1, Proposition 3.2). *Conditionally on four named statements (H1)–(H4) of the manuscript, for every viscosity $\nu > 0$ and all small τ : $\|\text{curl } u_\nu(1 - \tau)\|_2^2 \geq \nu^{3/2} C_\Omega \tau^{-1/2-3h}$ and $\|\partial_t u_\nu(1 - \tau)\|_2^2 \geq \nu^{5/2} C_u \tau^{-3/2-3h}$; so $\mathcal{K}_B \rightarrow \infty$ with finite integral and $\mathcal{K}_E \rightarrow \infty$ with infinite integral. Moreover $\|u(1 - \tau)\|_{L^3} \geq c \tau^{-4h/3}$, $\|\omega(1 - \tau)\|_\infty \geq c \tau^{-1-h}$ and $\tau^{1/2} \|u(1 - \tau)\|_\infty \geq c \tau^{-h}$: the blowup is of Type II.*

What the results mean

The radius theorem places the breakdown of the hydrodynamic gradient expansion of the Rindler fluid at a specific, certified event: the diffusive mode meets the first non-hydrodynamic mode at purely imaginary frequency and real momentum (Figure 1). This is the flat-cutoff instance of the convergence criterion of Grozdanov, Kovtun, Starinets and Tadić [10], with $\eta/s = 1/(4\pi)$, and the partner mode starts at the first Matsubara frequency of the cutoff's Unruh temperature. The conditional rates say how singular the manuscript's solutions are: the blowup exceeds the Type I rate by the angular Reynolds number τ^{-h} , and the classical necessary conditions for a singularity recalled in Section 3 hold with margins that are positive powers of τ^{-h} . For the map to Yang–Mills, the rates are exactly those the repository uses, and their status is conditional on named statements rather than unchecked.

The bridges

The workbench connects the Navier–Stokes construction to the author's other programmes through explicit maps: an abelian and a non-abelian map to Yang–Mills, the auxiliary torus operator that the Erdős–Straus supplement uses, Mellin transforms toward the zeta programme, a Beltrami solution from the S^6 oscillator, the divergence-free field of the Jacobian-conjecture counterexample, and the snapshot and Rindler maps to Einstein gravity. Section 5 states what each carries and what is established about it, with my assessment. In the author's view such connections are among the most valuable outcomes of the research. In my assessment the map to

Einstein gravity has produced the strongest result so far, the exact radius of Theorem I, because it settles a question the continuation left open and is of independent interest in fluid/gravity duality.

How this record was made

This record is an audit, carried out on 27 September 2026, of the workbench folder `navier-stokes/` of KokunoYumeto/`yang-mills-interacting-workbench` at commit `fa79faf`: the source-faithful transcription of the manuscript (cited *src p. N*), the 208-page workbench reader, and the continuation `continuations/20260919-vacuum-hydrodynamics/` (cited *RESEARCH.md*). The official PDF of the manuscript was checked against the hash that the transcription freezes (Appendix A). The construction is OpenAI’s; the reconstruction, the continuation and the maps are the workbench’s, whose author is KokunoYumeto.

The audit was carried out by Claude (Anthropic), model `claude-opus-5-5` (*claude-ab* below). A first-pass inventory was made by a separate Claude instance, and an independent referee, which wrote none of the text, re-verified the certificate of Theorem 2.2 with a second interval library and read the rest in both directions; its findings were verified by *claude-ab* before they were applied (Appendix A).

Every statement has one of four statuses: *verified* (proved here, with computer assistance where stated, or re-computed by a program listed in Appendix A); *not verified here*, which implies nothing in either direction; *my assessment*, a view in the first person with its reasons; and *proved negative*, stated with its counterexample or derivation (Section 4). Results that rest on statements of the manuscript are labelled Conditional and name them; they are verified relative to those statements. The manuscript’s main theorem is *not verified here*.

The audit note 51_ on which this paper is based was withdrawn from the branch `claude/claude-ab-grind-20260925` on 27 September 2026 because of its framing, together with the other notes. It remains, unchanged, in the branch’s history at commit `a90d9e5b`. The scripts cited here are republished in `contrib/claude-ab/rewrite/checks/ns/` on the same branch.

Organization

Section 1 states the manuscript’s theorem, the architecture of its proof and what the re-run of the workbench’s scripts verifies. Section 2 proves the collision, the exact radius and the coefficient asymptotics. Section 3 proves the conditional curvature rates and the Type II classification. Section 4 collects the proved limits, Section 5 the typed maps to the other programmes with my assessment, and Section 6 the questions and what was not checked. Appendix A lists the programs, and Appendix B what this typeset version changes relative to the audit note.

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1 The manuscript, the workbench, and what the re-run verifies

1.1 The imported theorem

The manuscript’s main theorem, which we call NS-00 as the workbench does, is the following [1, Theorem 1.1].

Theorem 1.1 (NS-00; the manuscript’s Theorem 1.1). *For every $\nu > 0$ there are a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty))$, a compact set K and a smooth solution (u, p) of the forced Navier–Stokes equations on $\mathbb{R}^3 \times [0, 1)$ with $u(\cdot, 0) = 0$, supports in K and $\sup_t \|u(t)\|_2 < \infty$, such that $\limsup_{t \uparrow 1} \|u(t)\|_\infty = \infty$.*

Status. Imported; *not verified here*. Corollary 10.6 of the manuscript (src p. 125) is the periodic version. The workbench is a reconstruction and verification reader of this theorem, and its own ledger lists four items as unfinished.

1.2 The architecture of the proof

The workbench’s reader follows the manuscript’s architecture, which we recall for orientation; none of these steps is verified here.

- *Force as residual* (src p. 3). For any incompressible (u, p) put $f := \partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p$. The work is to make f smooth through $t = 1$ while $\|u\|_\infty \rightarrow \infty$.
- *A slender core* (src p. 4). It has radial scale $\ell_r \asymp \tau^{1/2}$, axial scale $\ell_z \asymp \tau^{1/2-h}$ and speed $|u_\theta|, |u_z| \asymp \tau^{-1/2-h}$, with $\tau = 1 - t$ and $0 < h < 1/100$. The core energy is $\asymp \tau^{1/2-3h}$ and the angular Reynolds number is $\text{Re}_\theta \asymp \tau^{-h}$. The similarity coordinates are (src (3.2))

$$\tau = q(1 - \eta^2), \quad z = q^D \eta, \quad r^2 = 2qX, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h.$$

- *Profiles* (Theorem 4.6, src p. 32). The tangential residuals are exact radial divergences of a stress $q^{-A-1/2}T_0$, with $T_0 = 0$ on the inner region $X \leq X_a$ and admissible on the annulus; the exterior is an exact heat solution.
- *Base correction* (Proposition 5.5, src p. 60). The residual of the base field equals stress divergences plus a flat remainder. The profile estimates (5.42) hold uniformly on $[X_{\text{lo}}, X_{\text{hi}}] \times [-1, 1]$ for the summed background, with the Stokes streamfunctions summed before differentiation.
- *Waves and corrections* (§§6–9). Pulses on an auxiliary torus with the matrix $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ realise the stress through their covariance (Proposition 7.5); a four-step cycle raises the residual exponent $\sigma_j = \frac{1}{5} + \frac{j}{10} \rightarrow \infty$; a summation with shrinking cutoffs (Proposition 9.9) gives the manuscript’s Theorem 3.1.
- *The endpoint* (§10). Potential-first cutoffs (Proposition 10.1) are used; the force jets converge (Lemma 10.2) and are extended by a Borel series past $t = 1$ (Lemma 10.3); energy and uniqueness follow (Lemmas 10.4–10.5). The growth path is (10.20)–(10.21), the viscosity map is (10.22), and the periodisation is Corollary 10.6.

1.3 What the re-run of the workbench’s scripts verifies

The first-pass inventory classified the manuscript’s statements and the thirteen reconstruction components of the workbench. Its ten scripts were re-run here with python-flint 0.9.0, sympy 1.14.0 and mpmath 1.3.0, and their outputs agree with the shipped ones line for line, apart from timing lines. They comprise 251 identity checks and the 5 conditions of the collision certificate, all passing, and 4 negative controls, all correctly rejected. The workbench’s own replay, of 9 programs and 144 result groups, passes as well.

The identities verified in this way are:

- the leading-field calculus, including $R^{(0)} = -(\partial_r + k/r)(q^{-A-1/2}T_0)$ in full;
- the order- n coefficient equations for $n \leq 2$, from the full residual;
- the actual-shear correction (AS7)–(AS40) and the ambient projection;
- the vacuum-hydrodynamics identities: the transverse-traceless lift, the Lichnerowicz reduction, the momentum constraint, the Kasner bijection, the Einstein check $\text{Ric} = -4g/L^2$ and the Kretschmann limit;
- all bridge identities of Section 5;
- the viscosity, parabolic and periodisation maps.

Status. Verified (re-run).

Not verified here are NS-00 and its analytic estimates (the existence part of Theorem 4.6; Propositions 5.3, 5.5, 7.2–7.6, 9.6 and 9.9; Lemmas 10.2–10.5), the Volterra convergence and the contraction constants of the manuscript’s Appendix B, the stage-cycle exponent tables beyond the re-run of the bundle, and the continuation’s boundary relation (2.3) with its Brown–York formulas (20)–(24).

2 The Rindler shear sector: the collision and the exact radius

2.1 The dispersion relation

The vacuum-hydrodynamics continuation starts from the static Rindler metric

$$ds^2 = -(\rho/\rho_c)^2 c^2 dt^2 + d\rho^2 + dz^2 + dy^2 + dx_\perp^2, \quad 0 < \rho \leq \rho_c, \quad \rho_c = 2\nu/c,$$

whose cutoff surface $\rho = \rho_c$ is flat. For a shear perturbation $\propto e^{-i\omega t + ikz}$ the linearised vacuum equations reduce to the modified Bessel equation for a dual scalar φ , and the boundary condition at the cutoff, zero boundary source, gives $\varphi'(\rho_c) = 0$ (RESEARCH.md (10)–(16)). With $q = k\rho_c$, $w = \omega\rho_c/c$ and $\alpha = -iw$, the shear frequencies are therefore the zeros of

$$F(\alpha, q) := I'_\alpha(q), \quad (1)$$

the derivative of the modified Bessel function in its argument. These reductions are re-derived by the first-pass scripts.

Status. Verified (the first pass re-derived (13)–(14); (16) follows from (12) at $\rho = \rho_c$).

The statements of this section are theorems about the zeros of $\alpha \mapsto I'_\alpha(q)$; they use no input from the manuscript. Reading those zeros as the Rindler shear frequencies uses the continuation’s reduction.

A normalisation. Put $x = q^2/4$ and let $(1 + \alpha)_m$ be the rising factorial. The series $I_\alpha(q) = \sum_m (q/2)^{2m+\alpha} / (m! \Gamma(m + \alpha + 1))$ gives

$$qI'_\alpha(q) = \frac{(q/2)^\alpha}{\Gamma(1 + \alpha)} G(\alpha, x), \quad G(\alpha, x) = \sum_{m \geq 0} \frac{(\alpha + 2m) x^m}{m! (1 + \alpha)_m}. \quad (2)$$

For $|\alpha| < 1$ the prefactor is finite and nonzero, so $G = 0$ exactly when $F = 0$, and $G = G_\alpha = 0$ exactly when $F = F_\alpha = 0$. Since $G(\alpha, 0) = \alpha$ and $G_\alpha(0, 0) = 1$, the implicit function theorem gives a unique analytic root $\alpha_h(x)$ with $\alpha_h(0) = 0$: the *hydrodynamic root*. Its expansion begins $\alpha_h = -q^2/2 - 3q^4/16 - 29q^6/192 - \dots$, which in physical units is

$$\omega(k) = -ivk^2 - i\frac{3\nu^3}{2c^2}k^4 - i\frac{29\nu^5}{6c^4}k^6 - \dots$$

(RESEARCH.md (27)–(28)). The identity (2) was checked against mpmath’s Bessel functions by the referee at 60 random points with $|\alpha| \leq 0.95$, $|x| \leq 0.2$, agreeing to 10^{-60} .

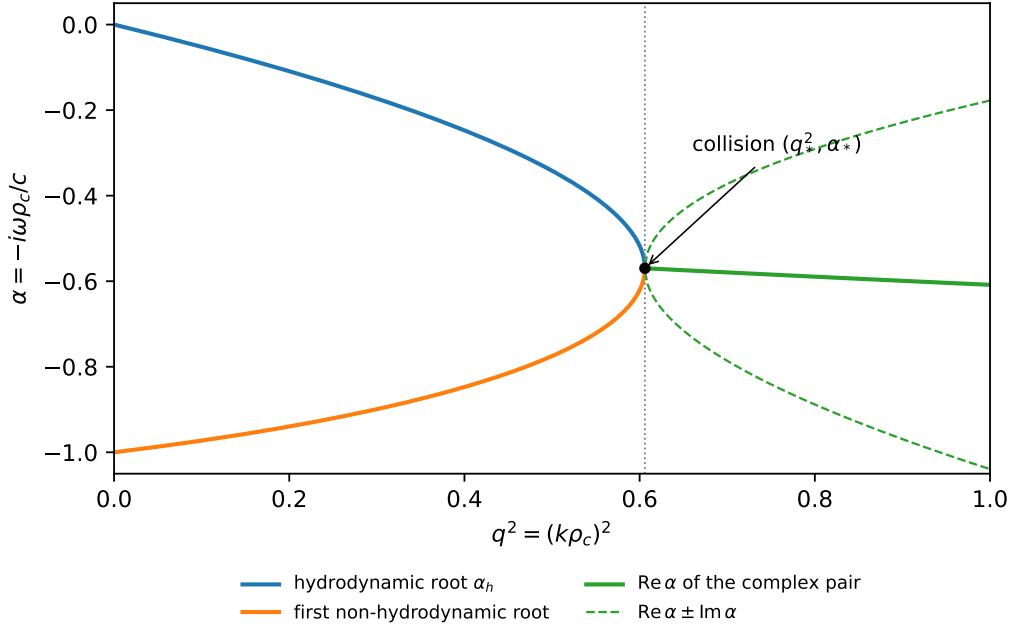


Figure 1: The real zeros of $\alpha \mapsto I'_\alpha(q)$ near the collision, computed with mpmath (the figure's script is in the paper's source folder). For $q^2 < q_*^2 = 0.60602 \dots$ the hydrodynamic root α_h and the first non-hydrodynamic root are real and distinct; they meet at (q_*^2, α_*) in a square-root fold (Theorem 2.1), and beyond it they form a complex-conjugate pair, shown by its real part and by $\text{Re } \alpha \pm \text{Im } \alpha$. The figure is an illustration; the statements are certified in Theorems 2.1 and 2.2.

The question the continuation left open is whether this series converges up to the first collision of the hydrodynamic root with another root, or whether a nearer complex singularity limits it. The answer is the first, and Figure 1 shows the two real roots meeting.

2.2 The collision

Theorem 2.1 (the collision). *There is (α_*, q_*) with*

$$\begin{aligned} |q_* - 0.778472800990330076180356446189| &< 10^{-20}, \\ |\alpha_* + 0.569714080972361784438457668663| &< 1.5 \cdot 10^{-10}, \end{aligned}$$

at which $F = F_\alpha = 0$. At that point $F_q \in 1.75035806 \pm 7.6 \cdot 10^{-9}$ and $F_{\alpha\alpha} \in 5.00 \pm 0.0203$, both positive. So the two roots meet in a square-root fold: they are real for $q < q_*$ near q_* and complex conjugate for $q > q_*$ near q_* .

Status. Proved here with computer assistance, in Arb ball arithmetic through python-flint. The program is `first_pass/certify_collision.py`; it was read line by line and re-run.

Proof. Let $I = a_0 \pm 1.5 \cdot 10^{-10}$ and $J = q_0 \pm 10^{-20}$ with the centres displayed above. Ball arithmetic shows:

- (i) $F > 0$ on $\partial I \times J$;
- (ii) $F(a_0, q_0 - 10^{-20}) < 0$;
- (iii) $F > 0$ on $I \times \{q_0 + 10^{-20}\}$.

The function $g(q) = \min_{\alpha \in I} F(\alpha, q)$ is continuous, with $g(q_0 - 10^{-20}) < 0 < g(q_0 + 10^{-20})$. At a zero q_* of g the minimiser α_* is interior to I by (i), so $F = F_\alpha = 0$ there. By the Bessel equation,

$F_q = I''_\alpha(q) = (1 + \alpha^2/q^2)I_\alpha - I'_\alpha/q$, which is enclosed directly; $F_{\alpha\alpha}$ is enclosed by a divided difference with a Cauchy remainder, the maximum modulus being taken on a circle of radius 0.05. Since $F_q \neq 0$ and $F_{\alpha\alpha} \neq 0$, the two roots split as a square root in $q - q_*$. $\square$

2.3 The radius, exactly

Theorem 2.2 (the radius of the hydrodynamic series). *Let $x_* = q_*^2/4$.*

- (a) *The hydrodynamic root α_h continues analytically to the open disk $|x| < x_*$. It is real and negative for $0 < x < x_*$, so the shear mode is purely damped there.*
- (b) *As $x \uparrow x_*$ along the reals, $\alpha_h(x) \rightarrow \alpha_*$, and α_h has a square-root branch point at x_* .*
- (c) *The Taylor series of α_h in x has radius of convergence exactly x_* . Equivalently, the series in q^2 has radius exactly $q_*^2 = 0.6060199018817300556 \pm 5.3 \cdot 10^{-20}$; in physical units $\omega(k)$ has radius exactly $k_* = q_* c/(2\nu) = 0.389236400495165038 \dots c/\nu$ in k , and its first singularity is $\omega_* = -0.2848570405 \dots i c^2/\nu$.*
- (d) *For real x slightly above x_* the two roots near α_* are complex conjugates: the modes propagate.*

Status. Proved here with computer assistance (`certify_hydro_radius.py`, certified in 137 s). Independently re-verified by the referee: all Krawczyk conditions with a second interval library (`mpmath.iv`, 36 terms, an independently derived tail bound), coverage and all consistency pairs in exact integer arithmetic, and the fold patch at twice the resolution.

Proof. Every step is a finite ball-arithmetic computation.

Rigorous evaluation. G and G_α are summed to $M = 30$ terms in Arb. For $|\alpha| \leq a = 0.95$ and $|x| \leq X = 0.2$ the terms satisfy $|t_m| \leq b_m := (a + 2m)X^m/(m! \prod_{k \leq m} (k - a))$ and $|\partial_\alpha t_m| \leq d_m := X^m/(m! \prod_{k \leq m} (k - a)) \cdot (1 + (a + 2m) \sum_{k \leq m} 1/(k - a))$. For $m \geq 30$, $b_{m+1}/b_m \leq 2X/((m+1)(m+1-a)) \leq \frac{1}{2}$ and $d_{m+1}/d_m \leq 5X/((m+1)(m+1-a)) \leq \frac{1}{2}$, so the tails are at most $2b_M$ and $2d_M$; they are added to the ball radii.

The Krawczyk tiling. The closed disk $|x| \leq R_{hi}$, with $R_{hi} \geq x_*$ the upper end of the certified box, minus the open square of half-width $3 \cdot 2^{-10}$ about a centre $X_c = x_* + O(3 \cdot 10^{-8})$, is covered by 42,650 square cells, subdivided adaptively 3×3 . For each cell X_i and a square α -box A_i with centre c and a real scalar Y , the Krawczyk operator

$$K_i = c - YG(c, X_i) + (1 - YG_\alpha(A_i, X_i))(A_i - c)$$

lies in the interior of A_i .

- *Existence.* A_i is convex and G is analytic in α . By the mean-value integral $G(\alpha, x) - G(c, x) = \int_0^1 G_\alpha(c + t(\alpha - c), x) dt \cdot (\alpha - c)$, whose integral lies in the convex enclosure of $G_\alpha(A_i, X_i)$, the map $N_x(\alpha) = \alpha - YG(\alpha, x)$ sends A_i into $K_i \subset A_i$ for every $x \in X_i$. Brouwer's theorem gives a fixed point, which is a zero because $Y \neq 0$.
- *Uniqueness and simplicity.* Two zeros in A_i , or a zero of G_α in A_i , would put 0 in the convex hull of $G_\alpha(A_i, x)$. Then $1 \in 1 - YG_\alpha(A_i, X_i)$, and K_i would contain the translate $c - YG(c, x) + (A_i - c)$ of A_i , which cannot lie in the interior of A_i .
- So for every $x \in X_i$ there is exactly one zero in A_i ; it is simple, lies in K_i and is analytic in x . The cells are enlarged by the factor $1 + 10^{-9}$, so that the rounding of cell centres leaves no gaps, and every wanted cell is certified or subdivided, with no failures.

Consistency. On all 168,301 touching pairs, $K_i \subseteq A_j$ or $K_j \subseteq A_i$, so the unique zeros agree on the overlaps. The cells containing $x = 0$ contain $\alpha = 0$. So the zeros form one analytic function on the tiled region, the continuation of α_h .

Realness. The 102 cells that meet the real axis have real centres and real-symmetric α -boxes, and uniqueness forces the zero to be real for real x .

The fold patch. Let A_F be the square of half-width $\frac{1}{4}$ about a dyadic $\alpha_0 \approx \alpha_*$, and X_F the square of half-width 2^{-8} about X_c ; all corners and boundary points are exact binary numbers.

(Fa) $G \neq 0$ on $\partial A_F \times X_F$, with $\min |G| \geq 0.0774$.

(Fb) The winding number of $G(\cdot, X_c)$ along ∂A_F is 2 (enclosure $2.00000 \pm 3 \cdot 10^{-10}$). So for every $x \in X_F$, $G(\cdot, x)$ has exactly two zeros β_1, β_2 in A_F , counted with multiplicity, and their discriminant $\Delta = (\beta_1 - \beta_2)^2 = 2p_2 - p_1^2$, with $p_k = (2\pi i)^{-1} \oint_{\partial A_F} \alpha^k G_\alpha / G d\alpha$, is analytic on X_F .

(Fc) On the 256 boundary segments of X_F the two zeros lie in disjoint Krawczyk boxes inside A_F , and the winding number of Δ along ∂X_F is 1 (enclosure $1.00000 \pm 3 \cdot 10^{-11}$). So Δ has exactly one zero in X_F , and it is simple. That zero is x_* , because Theorem 2.1 places the double zero (α_*, x_*) in $A_F \times X_F$.

(Fd) All 4,039 tiled cells that meet X_F have $K_i \subseteq A_F$, so there α_h is one of β_1, β_2 .

Conclusion (a). On $X_F \cap \{|x| < x_*\}$, a convex set with x_* on its boundary, $\Delta \neq 0$, so β_1 and β_2 are distinct and each is analytic there. The tiled function equals one of them on the connected set formed by the tiled region, X_F and the disk $\{|x| < x_*\}$: this set contains the part of the square annulus between the half-widths $3 \cdot 2^{-10}$ and 2^{-8} that lies inside the disk, and the circle $|x| = x_*$ passes through the central hole. So α_h is analytic on $|x| < x_*$. On the real segment $[X_c - 2^{-8}, x_*)$, α_h is real at a tiled point, so its partner is real too, Δ is real and nonzero, and it is positive at the tiled end; hence $\Delta > 0$ and both zeros are real up to x_* . Finally $G(0, x) = \sum_m 2m x^m / (m!)^2 > 0$ for $x > 0$, so $\alpha = 0$ is never a zero for $x > 0$; since α_h is real and continuous with $\alpha'_h(0) = -2$, it stays negative on $(0, x_*)$, and the frequency $w = i\alpha_h$ is negative imaginary.

Conclusions (b) and (d). Near x_* the two zeros are $(p_1 \pm \sqrt{\Delta})/2$ with a simple zero of Δ at x_* , and $\alpha_h \rightarrow p_1(x_*)/2 = \alpha_*$ as $x \uparrow x_*$. If α_h were analytic at x_* , then $\sqrt{\Delta} = \pm(2\alpha_h - p_1)$ would be analytic there, and Δ would have a zero of even order. For real $x > x_*$ near x_* , $\Delta < 0$, since the zero is simple and $\Delta > 0$ to its left; so the two zeros are a conjugate pair.

Conclusion (c). A power series at 0 with radius $R > x_*$ would agree with α_h on $|x| < x_*$ and be analytic at x_* , contradicting (b); the radius is at least x_* by (a). The physical constants follow from $k = q/\rho_c = qc/(2\nu)$ and $\omega = iac/\rho_c$. $\square$

The fold patch gives two further facts. The collision is the only branch point of the two roots in $A_F \times X_F$, and it is nondegenerate independently of the enclosures of F_q and $F_{\alpha\alpha}$ in Theorem 2.1. And the partner root is certified real, simple and distinct from α_h on $[X_c - 2^{-8}, x_*)$. As a negative control, the same tiling asked to cover the disk $|x| \leq 0.158 > x_*$ fails on the real axis beyond x_* , from $x = 0.15451$, as it must: there the two zeros are a conjugate pair, which no real-symmetric box can isolate.

2.4 The coefficients and the sum on the circle

Corollary 2.3 (coefficient asymptotics and boundary sum). *Write $\alpha_h(x) = \sum_{k \geq 1} a_k x^k$. Then*

$$a_k = -C x_*^{-k} k^{-3/2} (1 + O(1/k)), \quad C = \frac{\kappa \sqrt{x_*}}{2\sqrt{\pi}}, \quad \kappa^2 = \frac{4F_q}{q_* F_{\alpha\alpha}}.$$

The enclosures of Theorem 2.1 give $\kappa^2 \in [1.7915, 1.8061]$ and $C \in [0.14696, 0.14757]$; numerically $\kappa = 1.3403764770$ and $C = 0.1471754300$. In particular $a_k < 0$ for all sufficiently large k . The series converges absolutely on $|x| = x_$, and $\sum_k a_k x_*^k = \alpha_*$: the dispersion series summed at $k = k_*$ equals the collision frequency ω_* .*

Status. Proved here with computer assistance; the numerical value of C uses $F_{\alpha\alpha}$ to 30 digits (RESEARCH.md (32)). The sign $\kappa > 0$ is certified in ns51_sign_check.py.

Proof. A Δ -domain. Every closed tiled cell carries a simple zero, so by the implicit function theorem and compactness α_h extends to an open neighbourhood of the tiled region. The fold square X_F is a full neighbourhood of x_* , on which the two zeros are $(p_1 \pm \sqrt{\Delta})/2$ with $\Delta \neq 0$ off x_* . Hence α_h is analytic and single-valued on a slit disk $\{|x| < x_* + \varepsilon\} \setminus [x_*, x_* + \varepsilon)$ for some $\varepsilon > 0$, which contains a Δ -domain at x_* in the sense of singularity analysis.

The local expansion. Near x_* , $\alpha_h = (p_1 + \sqrt{\Delta})/2$ with p_1 and Δ analytic and $\Delta(x_*) = 0$ simple, and the quadratic model of F at the fold gives $\Delta = 4\kappa^2(x_* - x)(1 + O(x_* - x))$. Hence

$$\alpha_h = \alpha_* + \kappa\sqrt{x_*} (1 - x/x_*)^{1/2} + c_1(x_* - x) + c_2(x_* - x)^{3/2} + \dots$$

The sign $\kappa > 0$ holds because the hydrodynamic zero is the upper of the two real zeros on $[X_c - 2^{-8}, x_*)$. This is certified by a chain of 400 real Krawczyk cells that continues α_h from $\alpha = 0$ at $x = 0$ to $x_0 = X_c - 2^{-8}$, where it lies in $[-0.526, -0.474]$; the two fold zeros at x_0 are enclosed in $-0.484383876487 \pm 1.2 \cdot 10^{-13}$ and $-0.651760880396 \pm 4 \cdot 10^{-13}$, and the chain's enclosure meets only the upper one.

Transfer. By the transfer theorem of singularity analysis [9], $[z^k](1 - z)^{1/2} = -k^{-3/2}/(2\sqrt{\pi})(1 + O(1/k))$; the analytic term $c_1(x_* - x)$ contributes nothing for $k \geq 2$, and $(x_* - x)^{3/2}$ contributes $O(k^{-5/2})$. This gives the asymptotics, and $a_k x_*^k = O(k^{-3/2})$ gives absolute convergence on the circle. Abel's theorem and $\alpha_h(x) \rightarrow \alpha_*$ as $x \uparrow x_*$ give the boundary sum. $\square$

The first 240 Taylor coefficients were computed exactly as rationals, with $G(\alpha_h(x), x) = O(x^{241})$ checked exactly (hydro_series.py):

$$a_1 = -2, a_2 = -3, a_3 = -\frac{29}{3}, a_4 = -\frac{2843}{72}, a_5 = -\frac{392029}{2160}, a_6 = -\frac{14509367}{16200}, a_7 = -\frac{63074754607}{13608000}.$$

They reproduce the continuation's series in $q^2 = 4x$, including its fifth coefficient. All 240 are negative, and the referee extended this to 320. The Darboux-corrected ratios $(a_k/a_{k+1})/(1 + 3/(2k))$ approach $x_* = 0.1515049754 \dots$ with error $O(1/k^2)$: 0.15158 at $k = 40$, 0.151517 at $k = 100$ and 0.1515080 at $k = 200$. From the referee's 320 coefficients, $a_k x_*^k k^{3/2} \rightarrow -0.147175429969$ by Richardson extrapolation, which is $-C$ to 12 digits; the partial sum S_{320} plus the Darboux tail equals α_* to $1.6 \cdot 10^{-11}$; and the fold-patch discriminant has $\Delta'(x_*) = -7.186436 = -4\kappa^2$.

Status. Checked numerically.

2.5 The other modes, and the physical reading

The root that collides with α_h is the first non-hydrodynamic mode. Followed backwards in α from α_* (near the fold, q is a smooth function of α , since $F_q \neq 0$), it reaches $q \rightarrow 0$ as $\alpha \rightarrow -1$ with $x \approx 1 + \alpha$ (track_partner.py; numerical). As $q \rightarrow 0$ the zeros of $\alpha \mapsto I'_\alpha(q)$ tend to the zeros of $1/\Gamma(\alpha)$, namely $\alpha = 0, -1, -2, \dots$: indeed

$$2(q/2)^{1-\alpha} I'_\alpha(q) = \sum_m \frac{(2m + \alpha)(q/2)^{2m}}{m! \Gamma(m + \alpha + 1)} \rightarrow \frac{\alpha}{\Gamma(\alpha + 1)} = \frac{1}{\Gamma(\alpha)}$$

locally uniformly in α , and Hurwitz's theorem applies. (At $q = 0$ itself the boundary problem degenerates, so the statement is a limit.) The referee found numerically that $(\alpha + n)/q^{2n} \rightarrow (-1)^{n+1}n/((n!)^2 4^n)$ for $n = 1, 2, 3$. The cutoff observer has proper acceleration c^2/ρ_c and redshift factor 1 at $\rho = \rho_c$, so its Unruh temperature is $T = \hbar c/(2\pi k_B \rho_c)$, and the non-hydrodynamic frequencies tend to $\omega_n = -inc/\rho_c = -2\pi inT$ (units $\hbar = k_B = 1$): the Matsubara frequencies.

- *Normalisation.* With $2\pi T = c/\rho_c$, $(w, q) = (\omega, ck)/(2\pi T)$ are exactly the dimensionless variables $(\mathfrak{w}, q)$ of Grozdanov, Kovtun, Starinets and Tadić [10]. The collision is a critical point of the spectral curve $F(\alpha, q) = 0$ at real $q_*^2 = 0.6060199$ and imaginary $\mathfrak{w}_* = -0.5697141 i$, and Theorem 2.2 is the flat-cutoff instance of their statement that the radius of the hydrodynamic series is set by the nearest critical point of the spectral curve on the hydrodynamic sheet. Here that critical point lies at real momentum.
- *Shear viscosity.* $\rho_c = 2\nu/c$ gives $\nu = c^2/(4\pi T)$. With $\eta = \nu(\varepsilon + p)/c^2$ and $\varepsilon + p = Ts$ at zero chemical potential, $\eta/s = \nu T/c^2 = 1/(4\pi)$ in units $\hbar = k_B = 1$.
- *Pole collisions.* Areán, Davison, Goutéraux and Suzuki characterise the breakdown of diffusion near AdS_2 critical points by a collision of the diffusive pole with a non-hydrodynamic pole [2]; the same kind of event bounds the series here. At the collision, $|\omega_*|/(\nu k_*^2) = |\alpha_*|/(q_*^2/2) = 1.8802$, so at the radius the leading diffusive law is off by a factor 1.88.
- *The k^4 coefficient.* Compère, McFadden, Skenderis and Taylor write the corrected Rindler Navier–Stokes equation [6, (6.3)]; its linear part is $\partial_\tau v - r_c \partial^2 v = -\frac{3}{2} r_c^2 \partial^4 v$ in their time τ , with induced metric $-r_c d\tau^2 + dx^2$. In proper time $t = \sqrt{r_c} \tau$ and velocity $v/\sqrt{r_c}$ this becomes $\partial_t v - \sqrt{r_c} \partial^2 v = -\frac{3}{2} r_c^{3/2} \partial^4 v$, so $\nu = \sqrt{r_c}$ and $\omega = -i\nu k^2 - \frac{3}{2} i\nu^3 k^4$ ($c = 1$), which is the second coefficient above. (Only the linear terms of (6.3) were read, through a text extraction.) The k^6 coefficient and all higher ones come from the exact condition (1).
- *A two-pole model.* The telegraph model $w^2 + iw - q^2/2 = 0$, that is $\alpha^2 + \alpha + 2x = 0$, calibrated to the diffusion constant and to the first non-hydrodynamic mode, collides at $x = \frac{1}{8}$ ($q^2 = \frac{1}{2}$) and $\alpha = -\frac{1}{2}$, with hydrodynamic branch $-2x - 4x^2 - 16x^3 - \dots$. The exact problem moves the collision to $(q_*^2, \alpha_*) = (0.6060, -0.5697)$, with branch $-2x - 3x^2 - \frac{29}{3}x^3 - \dots$.

Status. The Unruh temperature, the normalisation and η/s are derived here; the comparisons with [10, 2] are readings of those papers’ abstracts and statements, not re-derivations.

A statement of the Rindler radius was not found in [4, 6, 14, 10, 17, 11]; that check was made through their abstracts and text excerpts, and the search was not exhaustive.

3 Curvature rates for the map to Yang–Mills

The Yang–Mills repository maps a velocity field u to the abelian connection $A_i = \lambda u_i T$ (the Yang–Mills paper [18], Section 6), whose curvature masses are

$$\mathcal{K}_B = \lambda^2 \|\text{curl } u\|^2, \quad \mathcal{K}_E = \lambda^2 c^{-2} \|\partial_t u\|^2,$$

and it derives lower rates for them from the manuscript’s profile estimates (ns_inner_profile_curvature_current_bridge.md, §§4–5, in yang-mills/sources/ym_gap_primary_20260908/). The rates are derived from named statements of the manuscript, and every step of the derivation was re-derived here, so their status is *conditional* on those statements.

3.1 The four statements used

All four are statements of the manuscript and are imported.

- (H1) Theorem 4.6(i) (src p. 33): $E = \sqrt{2X}F > 0$ for $X > 0$ and every $\eta \in [-1, 1]$, with $F = \varphi/C$ smooth on $[0, R] \times [-1, 1]$. In particular $E_0 > 0$ on the inner region, and $E_0(X, \eta) = O(\sqrt{X})$ at the axis.
- (H2) Proposition 5.5, (5.42) (src p. 60): for every composition $\mathcal{D}^I$ of $q\partial_q$, ∂_X and ∂_η , $\mathcal{D}^I(q^A u_{\theta, B} - E_0) = O(q^{2h})$ uniformly on $[X_{\text{lo}}, X_{\text{hi}}] \times [-1, 1]$. The source fixes radial endpoints $0 < X_{\text{lo}} < X_a <$

$X_b < X_{hi}$ and lets the constants depend on them; this is read as holding for each such choice, as the bridge file reads it. It holds for the summed background, with the Stokes streamfunctions summed before differentiation, so the summation cutoffs are already included.

- (H3) All annular corrections vanish on the fixed inner similarity region $X \in (0, X_a)$, for every η . Step 5 of Proposition 9.9 (src p. 116) says this at the chosen point X_{in} ; the region-wide statement is the sentence after (10.21) on src p. 124, and it is also carried by the support statement of Theorem 4.6(ii), by which the stress vanishes for $X \leq X_a$.
- (H4) Proposition 10.1: the cutoffs equal one on a neighbourhood of the singular point.

3.2 The rates

Theorem 3.1 (curvature rates). *Under (H1)–(H4), for every $\nu > 0$ there are $C_\Omega, C_{\dot{u}} > 0$ such that for all small $\tau = 1 - t$*

$$\|\operatorname{curl} u_\nu(1 - \tau)\|_2^2 \geq \nu^{3/2} C_\Omega \tau^{-1/2-3h}, \quad \|\partial_t u_\nu(1 - \tau)\|_2^2 \geq \nu^{5/2} C_{\dot{u}} \tau^{-3/2-3h}.$$

Hence, for the abelian image $A_i = \lambda u_{\nu,i} T$, $\mathcal{K}_B(1 - \tau) \rightarrow \infty$ while $\int_0^1 \mathcal{K}_B \leq \lambda^2 \mathcal{F}_\nu(1)^2 / (2\nu) < \infty$, and $\mathcal{K}_E(1 - \tau) \rightarrow \infty$ with $\int_0^1 \mathcal{K}_E = \infty$.

Status. Conditional on (H1)–(H4); every step re-derived, and every exponent computation checked symbolically, with three negative controls (ns51_rate_lemmas.py, 27 checks).

Proof. Choose $X_* \in (X_{lo}, X_a)$ with $e_0 = E_0(X_*, 0) > 0$, and $\eta_* > 0$ with $E_0(X_*, \eta) \geq e_0/2$ for $|\eta| \leq \eta_*$. By (H2)–(H4), $u_\theta \geq (e_0/4)q^{-A}$ on the circles $r = \sqrt{2X_*q}$, $|\eta| \leq \eta_*$, for small τ .

Enstrophy. On the disk D_z bounded by such a circle, Stokes' theorem gives $\int_{D_z} \omega_3 = 2\pi r u_\theta$, and Cauchy–Schwarz gives $\int_{D_z} \omega_3^2 \geq (2\pi r u_\theta)^2 / (\pi r^2) = 4\pi u_\theta^2$. At fixed τ , $dz/d\eta = \tau^D(1 - 2h\eta^2)(1 - \eta^2)^{-D-1}$, and integrating over $|\eta| \leq \eta_*$ gives the enstrophy bound with the exponent $D - 2A = -\frac{1}{2} - 3h$.

The time derivative. At a fixed point, $q_t = -1/L$, $\eta_t = D\eta/(qL)$ and $X_t = X/(qL)$ with $L = 1 - 2h\eta^2$. So $\partial_t u_\theta = q^{-A-1}(\mathcal{H} + O(q^{2h}))$ with $\mathcal{H} = (AE + XE_X + D\eta E_\eta)/L$. By (H1), $E = \sqrt{2X}F$ gives $\mathcal{H}(X, 0) = (1 + h)E(X, 0) + O(X^{3/2}) > 0$ for small $X > 0$. Integrating over a rectangle on which $|\mathcal{H}| \geq h_* > 0$, with the volume element $dx = q\tau^D L(1 - \eta^2)^{-D-1} dX d\theta d\eta$, gives the exponent $D - 2A - 1 = -\frac{3}{2} - 3h$.

Finiteness of $\int \mathcal{K}_B$. The energy inequality $\|u\|^2 + 2\nu \int \|\nabla u\|^2 \leq \mathcal{F}_\nu(t)^2$ and $\|\operatorname{curl} u\| = \|\nabla u\|$ for divergence-free, compactly supported fields give $\int_0^1 \mathcal{K}_B \leq \lambda^2 \mathcal{F}_\nu(1)^2 / (2\nu)$.

Viscosity scaling. The viscosity map (10.22) gives the factors $\nu^{3/2}$ and $\nu^{5/2}$. □

3.3 Type II blowup and the growth of the critical norm

Proposition 3.2 (Type II and critical-norm growth). *Under (H1)–(H4), for small τ :*

- (a) $\|u(1 - \tau)\|_p^p \geq c_p \tau^{3/2-h-p(1/2+h)}$ for every $1 \leq p < \infty$. The exponent is negative exactly for $p > p_h = (3 - 2h)/(1 + 2h)$, and $p_h < 3$. In particular $\|u(1 - \tau)\|_{L^3} \geq c \tau^{-4h/3} \rightarrow \infty$, and the lower bound diverges also for $p_h < p < 3$, although these norms are subcritical; for $p = 2$ the norm stays bounded by the energy inequality, and the core contribution decays like $\tau^{1/2-3h}$ (src p. 4).
- (b) $\|\omega(1 - \tau)\|_\infty \geq c \tau^{-1-h}$, from a disk mean away from the axis.
- (c) $\tau^{1/2}\|u(1 - \tau)\|_\infty \geq c \tau^{-h} \rightarrow \infty$. So the blowup is not of Type I, $|u| \leq C(T - t)^{-1/2}$: it is of Type II.

Status. Conditional on (H1)–(H4); the exponent algebra checked in ns51_rate_lemmas.py.

Proof. (a) Integrate $|u_\theta|^p \geq (e_0/4)^p q^{-pA}$ over a rectangle as in the proof of Theorem 3.1; the τ -exponent is $1 + D - pA$. For $p = 3$ the η -factor is $L(1 - \eta^2)^{-1+4h}$, which is integrable on compact η -sets. (b) The mean of ω_3 on the disk D_z is $2u_\theta/r \geq \sqrt{2}(e_0/4)X_*^{-1/2}q^{-1-h}$, and the maximum dominates the mean. The disk has radius $\sqrt{2X_*q}$, away from the axis, where (5.42) is not available, so no axis information is needed. (c) Use the growth path (10.21) with $q = \tau$. $\square$

Comparison with the classical necessary conditions for a singularity. Each of them is satisfied. The margins stated are lower bounds only, since no matching upper bounds are established here.

- *Leray's rates* [13]. A singular time forces $\|\nabla u(t)\|_2^2 \gtrsim (T-t)^{-1/2}$, and $\|u(t)\|_p \gtrsim (T-t)^{-(p-3)/(2p)}$ for $p > 3$; Robinson, Sadowski and Silva discuss these lower bounds [15]. The enstrophy exponent here is $\frac{1}{2} + 3h$, a margin of τ^{-3h} , and $\int \tau^{-1/2-3h} < \infty$ exactly when $h < \frac{1}{6}$, as the finite dissipation requires. For $p > 3$ the margin is $\tau^{-h(1+1/p)}$, and as $p \rightarrow \infty$ this becomes (c).
- *The L^3 criterion.* Escauriaza, Seregin and Šverák prove regularity under a bound in $L_t^\infty L_x^3$ for the unforced Cauchy problem [8], and Seregin shows that $\|u(t)\|_{L^3} \rightarrow \infty$ at a potential blowup time [16]. Here $\|u\|_3 \gtrsim \tau^{-4h/3}$ diverges. These criteria require divergence, not a rate, and whether they extend verbatim to smooth compactly supported forcing was not checked, so this is a statement of consistency, not of implication.
- *Axisymmetric Type I exclusion.* Chen, Strain, Tsai and Yau [5] and Koch, Nadirashvili, Seregin and Šverák [12] exclude Type I blowup for axisymmetric unforced solutions. The construction here is axisymmetric only on its inner region (H3), and the equation is forced, so those theorems do not apply as stated. Proposition 3.2(c) shows that it exceeds the Type I rate by at least the factor τ^{-h} , the angular Reynolds number of src p. 4.
- *$L_t^2 L_x^\infty$ and $\int \|\omega\|_\infty$.* $\int \|u\|_\infty^2 \gtrsim \int \tau^{-1-2h} = \infty$ and $\int \|\omega\|_\infty \gtrsim \int \tau^{-1-h} = \infty$, with margins τ^{-2h} and τ^{-h} against the scaling rates. These divergences hold for every singular solution, by Leray's bound $\|u(t)\|_\infty \gtrsim (T-t)^{-1/2}$ and by criteria of Beale–Kato–Majda type, so only the exponent arithmetic is claimed.

3.4 A formula of the growth path

The official PDF of the manuscript prints (10.20) as $x_\tau = (\sqrt{2X_{\text{in}}\tau}, 0, 0)$, with the radical over $2X_{\text{in}}\tau$ (checked on p. 124 of the PDF whose hash the transcription freezes). With this path, $q = \tau$ and $X = X_{\text{in}}$, and (10.21) follows from (3.6). The workbench's transcription renders the radical over the factor 2 only; on that path $X = X_{\text{in}}^2\tau \rightarrow 0$ and u_θ would grow only like τ^{-h} . So the source is correct, and the transcription's errata ledger, which was empty, gets this correction as its first entry.

Status. Verified against the official PDF.

4 Proved limits

Three statements of the preceding sections are limits, and each is proved.

- *The dispersion series does not converge beyond k_** (Theorem 2.2(c)). A series with a larger radius would make α_h analytic at x_* , where it has a square-root branch point. The negative control confirms this: the same tiling, asked to cover the disk of radius $0.158 > x_*$, fails on the real axis beyond x_* , where the two zeros are a conjugate pair.
- *Beyond the collision the shear modes propagate* (Theorem 2.2(d)). For real x slightly above x_* the two roots near α_* are complex conjugates, so the purely damped regime ends exactly at k_* .
- *The two-pole model misplaces the collision.* The telegraph model $\alpha^2 + \alpha + 2x = 0$, calibrated to the diffusion constant and to the first non-hydrodynamic mode, collides at $(q^2, \alpha) = (\frac{1}{2}, -\frac{1}{2})$, while

the exact collision is at $(0.6060, -0.5697)$. So the radius is a property of the exact dispersion relation and is not captured by the two-pole truncation.

The rates of Section 3 carry a limit of scope that the workbench states itself: the zero-quotient sequence of the map to Yang–Mills exists only along couplings $g_j \rightarrow 0$, and it rests on (H1)–(H4).

5 The typed maps to the other programmes

The workbench connects the Navier–Stokes construction to the author’s other programmes through explicit maps. The table states what each map carries and what is established about it here; *verified* marks an identity or a statement re-derived here.

Map	What it carries	Established here
NS $\rightarrow$ YM, abelian	$A_i = \lambda u_i T$; the curvature masses $\mathcal{K}_B = \lambda^2 \ \text{curl } u\ ^2$ and $\mathcal{K}_E = \lambda^2 c^{-2} \ \partial_t u\ ^2$	the map and its identities, verified; the divergence of the curvature masses conditional on (H1)–(H4) (Theorem 3.1); the zero-quotient diagonal needs couplings $g_j \rightarrow 0$
NS $\rightarrow$ YM, non-abelian	$\mathbf{A}_i = \lambda e_i \times u$; its fields, invariants and currents	all identities verified for every divergence-free u ; the off-axis vorticity bound of Proposition 3.2(b) under (H1)–(H4). The exact axis identity of the source’s (243) comes from the workbench’s own axis continuation, not from (H1)–(H4), since (5.42) does not reach the axis. The current is nonzero, so the image is a sourced field
NS $\rightarrow$ Erdős–Straus	the auxiliary torus operator $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$, with $\det J = 14$ and eigenvalues $4 \pm \sqrt{2}$	the operator identities used in 21 statements of the Erdős–Straus supplement, verified in the Erdős–Straus paper [19]
NS $\rightarrow$ zeta	angular-momentum balance, a Mellin shift and the Müntz multiplier $W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s})$	the transforms, verified; the growth, conditional on Theorem 3.1; the Mellin abscissa of $\mathcal{K}_B(1 - \tau)$ lies in $[\frac{1}{2} + 3h, 1]$, conditionally
$S^6 \rightarrow$ NS	a Beltrami-mode forced solution on $\mathbb{T}^3$ from the S^6 oscillator	a global smooth solution, verified exactly
Jacobian map $\rightarrow$ NS kinematics	$U = DF^{-1}(V \circ F)$, a divergence-free polynomial field whose flow escapes at $\tau = 1/8$	verified exactly (the Jacobian paper [20], §4)
NS $\rightarrow$ Einstein	the snapshot encoding, the Kasner map, the Rindler response and the slab comparator	all identities verified; the collision and the exact radius of Section 2

My assessment of these maps, taken together: the map to Einstein gravity has produced the strongest result so far, the exact radius of Section 2. I think it is of independent interest in fluid/gravity duality, because it places the breakdown of the hydrodynamic expansion of the Rindler fluid at a certified pole collision at real momentum, a question the continuation had left open. The maps to Yang–Mills are exact as identities, and their analytic content is conditional on (H1)–(H4); certifying those four statements (Question 3) would make the curvature rates unconditional. For the maps to the zeta programme and to the Erdős–Straus workbench, what is

established is the exact identities; what they carry about the zeta function or the equation beyond those identities is not verified here.

The connections to established work are these: the convergence of the hydrodynamic gradient expansion [10, 17, 11]; the breakdown of diffusion by pole collision [2]; fluid/gravity duality and the Rindler fluid [3, 4, 6, 14]; forced singularities, whose strategy of the force as a residual with controlled regularity the manuscript shares [7]; the classical blowup criteria of Section 3; and singularity analysis of power series [9]. The manuscript’s Lemma 10.3 is, in its structure, Borel’s lemma with shrinking cutoffs (a reading of the first pass), and its exterior heat profile $\mathcal{H}(Z) = Z^{-1-h}U(1+h, 2, 1/Z)$ is a Tricomi function (verified in `first_pass/check_axis_numerics.py`).

6 Questions, and what was not checked

6.1 Questions

1. Derive the continuation’s boundary relation (2.3) from the fluid/gravity linearisation of Bhattacharyya, Hubeny, Minwalla and Rangamani [3], and certify its Brown–York response formulas (20)–(24).
2. Certify the partner branch as real and simple on all of $0 < q < q_*$ by the same tiling, applied to a normalisation regular at $\alpha = -1$, such as $(1 + \alpha)G$ or F itself.
3. Certify the manuscript’s profile constants in interval arithmetic, and carry out the exponent bookkeeping of its correction cycle by hand.
4. Give explicit constants in the Lichnerowicz existence step of the continuation.
5. The manuscript needs $h > 0$ for its own steps. Can a construction of this kind work at $h = 0$, where it would sit exactly at the Type I rate?

6.2 What was not checked

- The manuscript’s analytic proof of NS-00, including whether (H1)–(H4) hold as stated.
- The continuation’s physics inputs: the relation (2.3), the Brown–York formulas (20)–(24), the Laurent data (38)–(41) and the two-sided probe; the slab theorem’s $H^2 \rightarrow H^s$ bootstrap; and the Lichnerowicz existence beyond the barrier inequalities.
- Whether the partner branch stays real and simple on all of $0 < q < q_*$; this is numerical, except on $[X_c - 2^{-8}, x_*)$, where it is certified (Section 2).
- The text of [6, (6.3)] beyond its linear terms.
- The literature search for prior statements of Theorem 2.2, which was not exhaustive.
- Two further discrepancies recorded by the first pass: a locator in the Yang–Mills repository (`ATTEMPTS.md`, line 59) points to the source bundle for four Navier–Stokes payloads that no bundle member contains; and the workbench reader’s statement of the leading-profile assembly theorem, which covers all of Theorem 4.6, is written but not independently validated, so Theorem 4.6 is treated as imported here.

A How this paper was checked

Sources. The Yang–Mills repository `KokunoYumeto/yang-mills-interacting-workbench` at commit `fa79faf`, folder `navier-stokes/`: the source-faithful transcription of the manuscript, the 208-page workbench reader and the continuation `continuations/20260919-vacuum-hydrodynamics/`. The official PDF of the manuscript [1] has SHA-256

0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f

and 2,959,204 bytes, equal to the hash that the transcription freezes; its p. 124 was read for (10.20).

The programs. They are republished, with their outputs, in `contrib/claude-ab/rewrite/checks/ns/` on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`.

- `first_pass/`: the ten scripts of the first-pass inventory (the leading-field calculus, 67 checks; the order- n equations, 10; the actual shear, 26; the cylindrical forms, 14; the axis numerics, 14; the vacuum hydrodynamics, 81; the bridges and scalings, 39; the collision certificate, 5 conditions; the branch tracking; the workbench replay), with their re-run outputs in `first_pass/rerun_by_claude_ab/`.
- `first_pass/certify_collision.py`: Theorem 2.1, in Arb ball arithmetic with python-flint.
- `certify_hydro_radius.py`: Theorem 2.2, certified in 137 s, with the negative control.
- `ns51_sign_check.py`: the sign $\kappa > 0$ of Corollary 2.3.
- `hydro_series.py`: the 240 exact coefficients.
- `track_partner.py`: the partner branch (numerical).
- `ns51_rate_lemmas.py`: the exponents of Theorem 3.1 and Proposition 3.2 (27 checks, 3 negative controls).
- `fig_collision.py`, in the paper’s source folder: Figure 1.

The referee pass. An independent Claude instance, which wrote none of the text, examined the audit note on which this paper is based, in both directions. It reported no major finding. It re-verified the certificate of Theorem 2.2 independently: all 42,650 Krawczyk conditions with a second interval library (`mpmath.iv`, 36 terms and an independently derived tail bound); coverage, the completeness of the neighbour search and all 168,301 consistency pairs in exact integer arithmetic; the fold patch at twice the resolution; and 320 coefficients by an Arb power-series Newton iteration. Its findings were verified by `claude-ab` before they were applied; among them are Corollary 2.3, with the Δ -domain, the Darboux asymptotics and the boundary sum; the uniqueness and nondegeneracy of the collision from (Fc); the certified realness of the partner near x_* ; the normalisation of [10] and $\eta/s = 1/(4\pi)$; the two-pole model; and the margins of Proposition 3.2. Two of its suggestions, a comparison with a specific relation of [2] and a further example from the literature, were not adopted, because they could not be checked against the papers. Its report and scripts are in `checks/ns51/referee/` in the branch history (below), and its cell dump is regenerated by `referee/certify_hydro_radius_rerun_dump.py`.

What has not been done. The manuscript’s main theorem is imported and not verified here, and Section 6 lists what else was not checked. This paper, as distinct from the audit note it is based on, has not had an independent referee pass.

B This version

This paper is the typeset version of the audit note 51_ of 27 September 2026 and of the condensed Markdown record written from it. The note was withdrawn from the branch on 27 September 2026, together with the other audit notes, because of their framing; it remains, unchanged, in the branch’s history at commit `a90d9e5b`, in `contrib/claude-ab/grind_20260925/`, with its checks in `checks/ns51/`.

- The mathematics of the note is kept: every theorem, proof, number and citation.

- The framing is corrected. The status classes of the note are mapped onto the four statuses of the introduction; the bridge table states what each map carries, instead of what it does not; remarks on the timing and the process of the audit are removed.
- Figure 1 is new. It illustrates Theorems 2.1 and 2.2 and was computed for this paper.

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